

Sample PDF Document: Vector Databases Overview

A vector database is a specialized database designed to store and search high-dimensional vectors efficiently. Unlike traditional databases that search by exact matches, vector databases search by similarity, finding the vectors that are closest in meaning to a given query vector.

Popular vector databases include FAISS (developed by Meta), Chroma, Pinecone, Weaviate, and Milvus. FAISS is a library rather than a full database, optimized for fast similarity search on large collections of vectors, and it runs entirely locally without needing a server.

Chroma is a lightweight, open-source vector database that is popular for local RAG projects because it is easy to set up, persists data to disk automatically, and integrates well with Python applications. It is often recommended as a first vector database to learn because of its simple API.

When choosing a vector database, factors to consider include the size of your dataset, whether you need it to run locally or in the cloud, how often your data changes, and whether you need metadata filtering alongside vector search.